INCH-POUND

MIL-C-44392(GL)
29 December 1989

MILITARY SPECIFICATION

COCOA BEVERAGE BAR, RATION LIGHTWEIGHT, 30-DAY

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cocoa beverage bar for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 8940

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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PPP-C-1752 - Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

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AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Ascorbyl palmitate. The ascorbyl palmitate shall comply with the Food Chemicals Codex.

3.2.2 Aspartame crystalline powder. Aspartame shall conform to the Food Chemicals Codex.

3.2.3 BHA solution. The BHA solution shall be a mixture of 30 percent BHA (food grade butylated hydroxyanisole) and 70 percent corn oil (see 6.4.7).

3.2.4 Cocoa powder, lecithinated. Cocoa powder shall be made from the nibs of domestically roasted, mature, well-fermented cocoa beans which have been properly dried. Either the nibs or the liquor from the nibs shall be mildly alkalized. The pH shall not be less than 6.0 nor more than 7.9. The cocoa powder shall contain from 10 to 12 percent cocoa butter. Chemically extracted cocoa, in whole or in part, shall not be acceptable. Not less than 99.0 percent of the powder shall pass through a U.S. Standard No. 100 sieve. The cocoa shall be salmonella negative in 125 grams.

3.2.5 Cream, sweet, spray-dried. Dry sweet cream shall comply with the FDA Standards of Identity for Dry Cream (21 CFR 131.149). The spray-dried sweet cream shall be of an off-white to cream color and have an odor and flavor characteristic of sweet cream. It shall contain 60 to 74 percent, by weight, of milkfat and shall contain no added nutritive sweeteners. The moisture content shall be not more than 2.0 percent. The spray-dried sweet cream shall be free from lumps that do not fall apart under light pressure, and 100 percent of the product shall pass through a U.S. Standard No. 12 sieve. It shall contain lecithin and an antioxidant to improve storage stability.

3.2.6 Cream substitute, non-dairy, dry. The creamer shall be a white, free flowing powder containing corn syrup solids, partially hydrogenated vegetable oils, sodium caseinate, mono and di-glycerides, dipotassium phosphate, artificial flavors, and colors.

3.2.7 Flavor, vanilla creme, artificial. The artificial vanilla creme flavor shall be in a powdered form, and shall have an odor and flavor typical of vanilla creme (see 6.4.1).

3.2.8 Flavor, vanilla, artificial. The artificial vanilla shall be in a powdered form, and shall have an odor and flavor typical of vanilla (see 6.4.2).

3.2.9 Lecithin. Lecithin shall comply with the requirements of the Food Chemicals Codex and shall be of the unbleached fluid type (see 6.4.3).

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3.2.10 Oil, sunflower. The sunflower oil shall possess a clean, bland flavor, be clear and brilliant at 70°F, and free from sediment and foreign material. The oil shall have a minimum stability of 15 hours (A.O.M.), a free fatty acid value not to exceed 0.05 percent, and shall pass a cold test of 5.5 hours minimum. The sunflower oil shall also have a linoleic acid content of 58.6 percent.

3.2.11 Pectin, extra-slow set. The extra-slow set pectin shall be highly methoxylated, and shall have a setting time between 260 to 330 seconds. The extra-slow set pectin shall comply with the requirements of the Food Chemicals Codex (see 6.4.4).

3.2.12 Pectin, rapid set. The rapid set pectin shall be very highly methoxylated, and shall have a setting time under 110 seconds, The rapid set pectin shall comply with the requirements of the Food Chemicals Codex (see 6.4.5).

3.2.13 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.14 Shortening, powdered. Powdered shortening shall be dried vegetable shortening on a non-fat milk carrier and may contain added emulsifiers. It shall have a fat content of 71 to 75 percent and shall be off-white in color (see 6.4.6). The moisture shall be not greater than 2.0 percent.

3.2.15 Silicon dioxide. The synthetic amorphous silicon shall comply with the Food Chemicals Codex.

3.2.16 Sugar, white, bakers special. Sugar shall be white, refined, granulated, bakers special cane or beet sugar, or a combination thereof.

3.2.17 Vitamin E. Vitamin E (dl-alpha-tocopherol) shall comply with the requirements of the Food Chemicals Codex, shall be in liquid form, and 1 mg shall be equivalent to 1.1 international units. Vitamin E shall have no off-odors, or off-flavors. Vitamin E acetate shall not be used.

3.2.18 Vitamin K1, spray dried. Vitamin K1 shall be spray dried phytonalione with 1 percent active vitamin K1.

3.2.19 D-calcium pantothenate. D-calcium pantothenate shall be the calcium salt of D-pantothenic acid and shall contain approximately 83 percent active pantothenic acid.

3.2.20 Pyridoxine hydrochloride, encapsulated and thiamine mononitrate, encapsulated. Both the pyridoxine hydrochloride and the thiamine mononitrate shall be encapsulated with approximately 66 percent mono- and di-glycerides and contain approximately 33 percent active pyridoxine hydrochloride and/or active thiamine mononitrate.

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3.2.21 Riboflavin 5' phosphate, sodium salt. The riboflavin 5' phosphate, sodium salt shall contain not less than the equivalent of 73 percent active riboflavin, B2.

3.2.22 Vitamin A acetate. The vitamin A acetate shall be in a stable, dry, dosage,beadlet form with an activity of at least 160 microgram RE activity per gram.

3.2.23 Vitamin B12, spray dried. Vitamin B12 (cyanocobalamin) shall be spray dried at 0.1 percent on a carrier of modified food starch with preservatives added.

3.2.24 Vitamin D2, ergocalciferol. The vitamin D2, USP, ergocalciferol shall be titrated with an inert dry carrier to contain approximately 40,000,000 units per gram. An appropriate stable, dry dosage form of vitamin D3 beadlets may be substituted for vitamin D2.

3.2.25 Water. Water used for sanitizing shall conform to the National Primary Drinking Water Regulations.

3.2.26 Nicotinic acid. The nicotinic acid (niacin) shall be a crystalline powder and shall conform to the Food Chemicals Codex or USP specifications.

3.2.27 Folic acid. The folic acid shall be a crystalline powder and shall conform to the Food Chemicals Codex or USP specifications.

3.2.28 d-Biotin. The d-Biotin shall be a crystalline powder and shall conform to the Food Chemicals Codex specifications.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Formulation of vitamin pre-mix. The vitamin pre-mix shall consist of the ingredients listed below. All ingredients shall be food grade, USP, FCC, or equivalent. The sugar shall have a granulation that aids in maintaining a stable mixture and all ingredients shall be free of hard agglomerates. Upon receipt by the contractor, the vitamin pre-mix shall be stored at a temperature in the range of 5° to 40°F, and shall be protected from light and moisture. When added to the cocoa bars, the vitamin pre-mix shall not impart an objectionable odor or flavor to the bars (see 6.4.8). The addition of vitamins shall comply with the Code of Federal Regulations (21 CFR 104.20).

Ingredient	Percent by weight
Nicotinic acid	3.530
Vitamin K_1, 1 percent spray dried	2.250
D-calcium pantothenate	1.960
Pyridoxine hydrochloride encapsulated	1.680
Thiamine mononitrate, encapsulated	1.170
Riboflavin 5' phosphate, sodium salt	0.930

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Ingredient	Percent by weight
Vitamin A acetate	0.700
Vitamin B_{12}, 0.1 percent spray dried	0.700
Folic acid	0.100
d-Biotin	0.030
Vitamin D_2, ergocalciferol	0.003
Sugar	86.947

3.3.2 Preparation and formulation of cocoa beverage base. The cocoa beverage base shall be formulated and prepared as follows:

Ingredient	Percent by weight
Cream, sweet, spray-dried	17.00
Sugar, bakers special	14.32
Cocoa powder, lecithinated	14.00
Cream substitute, non-diary, dry	29.00
Oil, sunflower	13.00
Shortening, powdered	7.00
Lecithin, fluid	1.50
Silicon dioxide	1.50
Vitamin pre-mix	1.00
Pectin, extra-slow set	0.75
Aspartame	0.27
Pectin, rapid set	0.25
Salt	0.25
Ascorbyl palmitate	0.06
Artificial vanilla creme flavor	0.04
Vitamin E	0.025
Artificial vanilla flavor	0.021
BHA solution	0.014

a. Blend sugar, non-dairy cream substitute, silicon dioxide, vitamin pre-mix, aspartame, pectins, salt, ascorbyl palmitate, and vanilla flavorings in an elliptical mixer on low speed for 10 minutes.

b. Add cocoa powder to above and blend for 3 to 5 minutes on low speed.

c. Add spray-dried cream and powdered shortening and blend for 3 to 5 minutes.

d. Combine oil, BHA solution, vitamin E and lecithin. Add to mixture and blend for approximately 2 minutes.

3.4 Compression of beverage bars. The cocoa beverage base mixture shall be compressed so as to produce bars complying with the finished product requirements (see 3.6).

NOTE: Satisfactory product has been compressed at pressures ranging from 180 to 400 psi.

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3.5 Pouch filling and sealing. After compression the product shall be packaged immediately in accordance with 5.1.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

Compressed product.

a. There shall be no foreign material such as, but not limited to, dirt, glass, wood, insect parts, hair, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. Each bar shall be 1.0 $\pm$ 0.1 inches wide by 3.0 $\pm$ 0.1 inches long by 0.9 $\pm$ 0.1 inch thick.

e. The weight of each bar shall be not less than 42 grams and not greater than 50 grams.

f. There shall be no compression streaks or stains.

g. No pouch shall contain a bar which has been broken or crumbled (more than three pieces).

h. The shape of the bars shall not be distorted.

i. The moisture content shall be not greater than 3.0 percent.

j. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

k. The E. coli count shall be less than 3 per gram (no positives in the standard three tube MPN technique) in four of the five samples and not greater than 20 per gram in any sample.

l. Each bar shall contain not less than the following amounts of vitamins (see 4.5.13):

Vitamin E	14.0 mg
Niacin	8.0 mg
Pantothenic acid	4.0 mg
Pyridoxine	1.5 mg
Vitamin B_{12}	1.8 mg
Riboflavin	1.0 mg
Thiamine	1.0 mg
Vitamin A	400 mcg RE

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Folic acid	250 mcg
Biothin	90 mcg
Vitamin K	60 mcg
Vitamin D	6 mcg

Rehydrated product.

m. The beverage shall have a color, odor, and flavor characteristic of sweetened cocoa.

n. When the product is rehydrated as specified in 4.5.6, the product shall show not more than 25 percent undissolved material.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food", and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any

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failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists, and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Laminated pouch material certification. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements.

Material requirement	Requirement paragraph	Test procedure
Thickness of films	5.1.1.1.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Laminated material identification and construction	5.1.1.1.1 and 5.1.1.2.1	Laboratory evaluation.

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Total thickness	5.1.1.2.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.

4.5.1.3 Unfilled pouch certification. A certificate of compliance may be accepted as evidence that unfilled pouches conform to requirements specified in 5.1.1.1.1 and 5.1.1.2.1. When deemed necessary by the USDA, testing of the unfilled preformed pouches for seal strength and integrity and air tightness of manufacturer's seals shall be as specified in 4.5.1.3.1 and 4.5.1.3.2.

4.5.1.3.1 Unfilled preformed pouch seal strength testing. The unfilled pouches shall be tested for seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The sample size shall be the number of pouches indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength of each seal shall be calculated by averaging the test results of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement or any seal failing to meet the average seal strength requirement specified in 5.1.1.1.1 or 5.1.1.2.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.1.3.2 Unfilled preformed pouch integrity and air tightness of manufacturer's seals testing. Pouch integrity and air tightness shall be determined by placing rigid material similar in size and shape to the cocoa beverage bar(s) into the pouch and sealing under a vacuum of not less than 25 inches of mercury. Not less than 96 hours after vacuum sealing, the pouch shall be examined for vacuum loss in accordance with the procedure in 4.5.4. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.2.1 In-process examination at the assembly plant. Bulked-packed bars shall be examined before unit packing into individual pouches. The lot size shall be expressed in shipping containers. The sample shall be the number of

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shipping containers indicated by inspection level S-3. Two bars shall be randomly selected from each of the shipping containers in the sample. Any bar found to have objectionable taste or odor (for example, burnt, scorched, stale, sour, rancid, or moldy) shall be cause for rejection of the lot.

4.5.3 Filled and sealed pouch examination (applicable to unit packed and bulk packed pouches). The filled and sealed pouches shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

TABLE I. Filled and sealed pouch defects (applicable to unit packed and bulk packed pouches) 1/

Category		Defect
Major	Minor	
101		Pouch size not as specified 2/
102		Not heat sealed as specified
103		Not clean 3/
104		Tear, hole, or open seal
105		Pouch has foreign odor
106		Evidence of delamination 4/
107		Seal widths not as specified
108		Closure seal located less than 1/8 inch above tear notch 5/
109		Presence of foldover wrinkles, occluded matter, and entrapped moisture or grease that reduces the effective closure seal to less than 1/8 inch 6/
110		Required labeling or marking missing, incorrect illegible, or that smudges

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TABLE I. Filled and sealed pouch defects (applicable to unit packed and bulk packed pouches) 1/

Category		Defect
Major	Minor	
	201	Tear notch missing 5
	202	Tear notch not located as specified 5/
	203	Depth of tear notch not as specified 5/

1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

2/ Measurement may be accomplished on unfilled pouches being used for current production lot.

3/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings, etc.) or generally detracts from the clean appearance of the pouch. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the pouch or by gently brushing the pouch with a clean dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

4/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Pouches exhibiting this type of delamination shall be tested by manually flexing the delaminated area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delamination area shall then be rapidly flexed by rotating both hands in alternating clockwise - counter-clockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between the thumb and forefinger, a number two stylus shall be inserted into the delaminated area and gentle lifting force applied against the outer ply. If separation of the outer ply

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can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, it shall be considered evidence of delamination.

5/ Applicable to unit packed pouches only.

6/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/8 inch wide from side seal to side seal that produces a hermetically sealed pouch.

4.5.4 Pouch vacuum examination. Not less than 96 hours after filling and sealing, the pouches shall be visually examined for conformance to the vacuum requirements in 5.1.1.1.2 and 5.1.1.2.2. The sealed pouch shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.5 Net weight examination of product when unit packed in pouches as specified in 5.1.1.1. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 42 grams or greater than 50 grams shall be classified as a minor defect. The lot size shall be expressed in pouches. The results shall be reported to the nearest gram. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.6 Product examination when unit packed as specified in 5.1.1.1. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE II. Product defects (unit pack) 1/ 2/

Category		Defect
Major	Minor	
		Compressed product
101		Dimensions of bar not as specified
102		Presence of compression stains or streaks
103		Color of bar not as specified

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TABLE II. Product defects (unit pack) 1/ 2/

Category		Defect
Major	Minor	
104		Odor and flavor of bar not as specified
	201	Broken or crumbled bar (more than three pieces) 3/
		Rehydrated product
105		After dissolving in water according to instructions on the pouch, the beverage does not have a color, odor, and flavor characteristic of sweetened cocoa
106		Product shows more than 25 percent undissolved material 4/

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by USDA agent).

3/ A broken or crumbled bar is a bar which either:

a. Has been broken into more than three pieces, the smallest of which is larger than 1/2 cubic inch in volume; or

b. Has more than 1/8 of the volume of the bar crushed.

4/ The percentage of undissolved material shall be determined as (specified in 4.6.1).

4.5.7 Product testing when unit packs are held over 30 days (see 5.1.1.1). When unit packed bars are held over 30 days before final assembly, the storage container shall be tested for conformance to the oxygen content or vacuum requirements of 5.1.1.1. Testing shall be as specified in 4.5.7.1 or 4.5.7.2. The lot size shall be expressed in number of storage containers. The sample unit shall be one sealed container. The inspection level shall be S-2. Any storage container failing to meet the oxygen content or vacuum requirements shall be cause for rejection of the lot.

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ll contain not less than

4.5.7.1 Oxygen content testing. The analysis to determine the percentage of oxygen in the nitrogen atmosphere shall be in accordance with any accepted USDA method, using appropriate methods to adapt to the bulk container in which the bars are stored.

4.5.7.2 Vacuum testing. The vacuum level shall be determined by testing in accordance with any accepted USDA method, using appropriate methods to adapt to the bulk container in which the bars are stored.

4.5.8 Examination for weight of product when bulk packed in pouches as specified in 5.1.1.2. The weight of the cocoa beverage bars shall be determined by individually weighing two bars randomly selected from each sample unit. Any individual bar weight of less than 42 grams or greater than 50 grams shall be classified as a minor defect. The lot size shall be expressed in bulk packed pouches. The results shall be reported to the nearest gram. The sample unit shall be one bulk packed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.9 Product examination when bulk packed as specified in 5.1.1.2. The finished product shall be examined for defects listed in table III. The lot size shall be expressed in bulk packed pouches. The sample unit shall be two bars selected at random from the bulk packed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE III. Product defects (bulk pack) 1/ 2/

Category		Defect
Major	Minor	
101		Dimensions of bars not as specified
102		Presence of compression stains or streaks
103		Color of bars not as specified
104		Odor and flavor of bars not as specified
	201	Broken or crumbled bar (more than three pieces) 4/
105		Number of bars not as specified 3/

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TABLE III. Product defects (bulk pack) 1/ 2/

Category		Defect
Major	Minor	
		Rehydrated product
106		After dissolving in water according to instructions on the pouch, the beverage does not have a color, odor, and flavor characteristic of sweetened cocoa
107		Product shows more than 25 percent undissolved material 5/

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by USDA agent).

3/ The sample unit shall be the selected bulk packed pouch.

4/ A broken or crumbled bar is a bar which either:

a. Has been broken into more than three pieces, the smallest of which is larger than 1/2 cubic inch in volume; or

b. Has more than 1/8 of the volume of the bar crushed.

5/ The percentage of undissolved material shall be determined as specified in 4.6.1.

4.5.10 Pouch closure seal strength testing. The filled and sealed pouches shall be tested for closure seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 inch in width and to a length suitable for proper mounting. Three adjacent specimens shall be cut from the closure seal of each pouch in the sample. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from the seal. The results shall be reported to the nearest 0.1 pound per inch of width. Any test specimen or average seal strength failing to meet the requirements of 5.1.1.1.2 or 5.1.1.2.2 shall be classified as a major defect and shall

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be cause for rejection of the lot. The lot size shall be expressed in pouches. The samle unit shall be one filled and sealed pouch. The inspection level shall be S-1.

4.5.11 Moisture content testing. The product shall be tested for moisture content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Sugar and Sugar Products; Section: Moisture Method: Drying in Vacuum. Test results shall be reported to the nearest 0.1 percent. Any test failure shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed unit packed pouch or two bars selected at random from bulk packed pouches. The inspection level shall be S-2.

4.5.12 Microbiological testing. Five filled and sealed unit packed pouches or two bars selected from each of five filled and sealed bulk packed pouches shall be selected at random from the lot regardless of lot size. The pouches or two bar units shall be individually tested for microbiological levels in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Microbiological Methods; Section: Frozen, Chilled, Precooked, or Prepared Foods - Official First Action. The diluent shall be added to each bar and allowed to stand for 15 minutes before blending the sample. Continue with the procedure as directed in the Official First Action. Any result not conforming to the microbiological requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.13 Vitamin content testing. Five bars of the product shall be selected at random from the lot regardless of lot size. Analyses of vitamins shall be conducted by methods approved by the AOAC or by appropriate chromatographic methods. Some modifications of extraction techniques may be required for accurate analyses of the vitamins.

4.5.14 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Cushioning material missing or not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Number of bars not as specified.
Pouches not packed flat in layers.

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4.6 Methods of inspection.

4.6.1 Procedure for determining percentage of undissolved material.

Equipment required:

a. 5-inch variable speed magnetic stirrer.
b. 1-1/2 inch teflon coated stirring bar.
c. U.S. Standard No. 10 sieve.
d. 1000 mL beaker.

Procedure:

a. Tare the sieve.
b. Crumble the cocoa beverage bar into very fine particles into the 1000 ml beaker.
c. Add 250 ml hot water (150° to 170°F) to the beaker. Add the magnetic stirring bar and turn on the stirrer to achieve a vortex.
d. Add the pieces of cocoa beverage bar into the vortex and mix for 5 minutes.
e. Immediately pour the contents of the beaker onto the sieve. Remove the stirring bar and shake excess water from the sieve.
f. Weigh the material remaining on the sieve.
g. Subtract the tare weight to obtain weight of undissolved material.
h. Calculate the residue as follows:

$$\frac{\text{Weight of undissolved material}}{\text{Weight of beverage bar}} \times 100 = \text{Percent of undissolved material}$$

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. The product shall be unit packed in accordance with 5.1.1.1 or bulk packed in accordance with 5.1.1.2.

5.1.1.1 Unit packs. When specified (see 6.1), one cocoa beverage bar shall be unit packed in a preformed pouch as specified in 5.1.1.1.1. When the bars are unit packed, they shall be stored in a closed container. If bars are held for more than 30 days from date of unit pack to date of assembly, they shall be protected from oxygen by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen or under a vacuum of at least 27 inches of mercury during the period exceeding 30 days. In no case shall the bars be held for more than 90 days from date of unit pack to ration assembly.

5.1.1.1.1 Preformed pouch for unit pack. The preformed pouch shall be fabricated from inside to outside 0.004 inch thick (± 0.001 inch) polyolefin film laminated to 0.0005 inch thick (± 0.0001 inch) polyester. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to

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the product being packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches. The preformed pouch shall be a flat style pouch having inside dimensions of 4-1/8 inches wide by 6-3/8 inches long (± 1/16 inch), with the 4-1/8 inch dimension being the opening of the pouch. The pouch shall be made by heat sealing three edges with 1/4 inch (± 1/16 inch) wide seals. The heat seals shall be made in a manner that will ensure hermetic seals. The pouch shall maintain its integrity and air tightness of the side and bottom seals when tested as specified in 4.5.1.3.2. The side and bottom seals shall have an average seal strength of not less than 10 pounds per inch and no individual specimen shall have a seal strength of less than 9 pounds per inch when tested as specified in 4.5.1.3.1. A tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch shall be provided with an extended lip, extending 1/8 inch (± 1/16 inch) to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended lip.

5.1.1.1.2 Filling and sealing of unit pack pouches. One beverage bar shall be placed into a preformed pouch as specified in 5.1.1.1.1. The filled pouch shall then be hermetically sealed by evacuating the air within a sealing chamber under a vacuum of not less than 25 inches of mercury, and applying a continuous heat seal of not less than 1/8 inch wide. The closure seal shall be free of foldover wrinkles, delamination, occluded matter, and entrapped moisture or grease. The closure seal shall be located not less than 1/8 inch above the tear notch. The average closure seal strength shall be not less than 10 pounds per inch and no individual test specimen shall be less than 9 pounds per inch when tested as specified in 4.5.10. The filled and sealed pouch shall show no loss of vacuum when examined as specified in 4.5.4.

5.1.1.2 Bulk packs. When specified (see 6.1), thirty beverage bars shall be bulk packed in a pouch as specified in 5.1.1.2.1. The bulk packed bars shall be held for not more than 180 days from date of pack to ration assembly.

5.1.1.2.1 Preformed pouch for bulk pack. The preformed pouch shall be fabricated from a hot fill type, minimum 0.006 inch thick, 3-ply laminate from inside to outside of polyolefin/aluminum foil/polyester material. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches. The preformed pouch shall be a flat style pouch having outside dimensions of 12 inches wide by 15 inches long (± 1/8 inch), with the 12 inch dimension being the opening of the pouch. The pouch shall be made by heat sealing three edges with 3/8 inch (± 1/8 inch) wide seals. The heat seals shall be made in a manner that will ensure hermetic seals. The pouch shall maintain its integrity and air tightness of the side and bottom seals when tested as specified in 4.5.1.3.2. The side and bottom seals shall have an average seal strength of not less than 10 pounds per inch and no individual specimen shall have a seal strength of less than 9 pounds per inch when tested as specified in 4.5.1.3.1.

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5.1.1.2.2 Filling and sealing of bulk pack pouches. Thirty beverage bars shall be single tier packed into the bulk pack pouch. The filled pouch shall then be hermetically sealed by evacuating the air within a sealing chamber under a vacuum of not less than 25 inches of mercury, and applying a continuous heat seal of not less than 1/8 inch in width. The closure seal shall be free of foldover wrinkles, delamination, occluded matter, and entrapped moisture or grease. The average closure seal strength shall be not less than 10 pounds per inch and no individual test specimen shall be less than 9 pounds per inch when tested as specified in 4.5.10. The filled and sealed pouch shall show no loss of vacuum when tested as specified in 4.5.4.

5.2 Packing for shipment to ration assembler. Two hundred and ten packaged bars, preserved as specified in 5.1, shall be packed flat in layers in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF, variety SW, grade 275, class domestic of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to type II, class B of PPP-C-843 or cushioning material not less than 1/16 inch thick, conforming to type I of PPP-C-1797 or cushioning material conforming to type VII, class 1 of PPP-C-1752 shall be placed between each of the layers and at the bottom and top of the box to minimize movement of the individual packs.

5.3 Labeling and marking.

5.3.1 Unit pack pouches. Unit pack pouches shall be labeled in a permanent, contrasting color directly on the packaging material in a manner that does not damage the pouches, with a food compatible ink which is free of carcinogenic elements or ingredients. No marking shall be closer than 1/16 inch to any seal. Pressure-sensitive labels may be used instead of marking directly on the packaging material. Labels shall be of a water-resistant grade of paper or film, flat-back, coated on one side with pressure-sensitive adhesive. The texture of the material shall permit flexibility. Labels shall be suitable for printing or writing on with ink without feathering or spreading. The adhesive shall be of a pressure-sensitive permanent type. The adhesive shall be water-insoluble, homogenous, and shall be coated in a smooth layer on one side of the label. The adhesive shall require no solvent, heat, or other preparation prior to application. Adhesive shall be of a type that will adhere to the pouch surface under high or low temperatures. Dimensions of the label shall not be larger than the dimensions of the surface of the product where the label is applied on the package. Ink and adhesive used must be compatible for use with food products and shall be free of carcinogenic elements or ingredients. Letter size shall be not less than 1/16 inch high. The following information must be included:

Item Name

Mixing Instructions: Mixing instructions for beverage bar shall be as follows:

Eat as is, or crumble into fine particles in canteen cup. Add 6 ozs. of water (1/3 canteen cup). Stir until dissolved.

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Date of production/lot number (Julian date) 1/

Name of contractor

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1987 would be coded as 7078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code.

5.3.2 Bulk pack pouches. Bulk pack pouches shall be marked as specified in 5.3.1, except that the labels shall include the following additional information:

a. Number of bars
b. Contract number
c. Name, address and zip code of contractor

5.3.3 Marking of shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
e. Whether bars are to be unit packed or bulk packed (see 5.1.1.1 and 5.1.1.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Ingredient information.

6.4.1 Flavor, vanilla creme, artificial. It has been found that artificial vanilla No. 610989U manufactured by Hercules, Inc. meets the requirements of 3.2.7 and performs satisfactorily in this product.

6.4.2 Flavor, vanilla, artificial. It has been found that artificial vanilla No. 650204U manufactured by Hercules, Inc. meets the requirements of 3.2.8 and performs satisfactorily in this product.

6.4.3 Lecithin. It has been found that fluid, unbleached lecithin Central 3F-UB manufactured by Central Soya Company meets the requirements of 3.2.9 and performs satisfactorily in this product.

6.4.4 Pectin, extra slow-set. It has been found that extra slow-set Mexpectin XSS100 manufactured by Grinstead Products, Inc. meets the requirements of 3.2.11 and performs satisfactorily in this product.

6.4.5 Pectin, rapid-set. It has been found that rapid-set Mexpectin RS461 manufactured by Grinstead Products, Inc. meets the requirements of 3.2.12 and performs satisfactorily in this product.

6.4.6 Shortening, powdered. It has been found that powdered shortening Beatreme S manufactured by Beatrice Food Ingredients, Inc. meets the requirements of 3.2.14 and performs satisfactorily in this product.

6.4.7 BHA solution. It has been found that 30 percent BHA in corn oil Tenox 4A manufactured by Eastman Chemicals meets the requirements of 3.2.3 and performs satisfactorily in this product.

6.4.8 Vitamin pre-mix. The following products have been used to produce a vitamin pre-mix in the laboratory, although other products may give equivalent results: Nicotinic acid, D-biotin, D-pantothenic, hemi calcium salt, and Vitamin D_2 (ergocalciferol, crystalline 40,000 i.u. 1 g) from Sigma Chemical Company, St. Louis, MO; thiamine mononitrate and encapsulated pyridoxine from Desmo Chemical Company, St. Louis, MO; Vitamin B_{12}, 0.1 percent spray dried, from Roche Chemical Division, Hoffman LaRoche Company, Nutley, NJ: folic acid from MCB/EM Science, Gibbstown, NJ Riboflavin 5' phosphate from E.M. Chemical Division of Merck Chemical Company, Hawthorne, NY: Vitamin A acetate crystalline beadlets with activity of 546,000 I.U./gram from Chemical Dynamics Company of South Plainfield, NJ; and Vitamin K1 (phytonadione) as one percent spray dried from Danochem USA., Inc., Hoffman LaRoche Company, Nutley, NJ.

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6.5 Subject term (key word) listing.

Combat field feeding
Operational ration

Custodian:

Army - GL

Review activities:

Army - MD, QM
Navy - MC, SA
Air Force - 50
DP - SS

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44392(GL)	COCOA BEVERAGE BAR, RATION LIGHTWEIGHT, 30-DAY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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